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1 Introduction

BrickIt is an arcade game, based on the Arkanoid and Breakout game series. The player controls a pad that he uses to bounce a ball to clear all the bricks in the game area.

BrickIt is developed by Thijs van As. He can be reached at t.vanas@gmail.com. Please feel free to ask any questions.

2 Features

2.1 Game Features

BrickIt has the following features:

- 20 levels (very easy to extend with more)
- 7 sound effects
 - Sound can be enabled/disabled in menu
 - Sound can be deactivated overall (to save loading times)
 - Can be changed to other sound effects with minor 'code' change
- Main menu including Instructions and Level Select
- In-game pause button (also provides return to main menu feature)
 - Auto-pause on rotation of mobile device
 - Auto-pause on loss of focus in PC browser (for example switching tab)

- Use 'Escape' to pause with keyboard
- Fully playable and navigable using multiple input devices (Section 2.3)
 - Touch screen
 - Tap or swipe (only enabled for iPad)
 - Keyboard
 - Mouse
- Playable (and tested) on many devices (Section 3)
 - Auto-detection of different devices:
 - iPhone/iPod Touch
 - iPad
 - Android
 - PC
- 100% vector graphics: image quality perfect on all devices/screen sizes
- Offline caching for mobile devices: the game can be played without network connectivity

2.2 Special Editions

For Apple iOS, a special web app version is available for BrickIt. When a player loads the game using an iPhone or iPod touch, he will be asked to install the app to his home screen.

iOS 3.2, currently shipped on the iPad, has a serious bug with Canvas Text - support for this is broken. This results in the fact that many parts of the game were not usable, mainly the menus. So the option was added to disable the menus. This option is automatically enabled when an iPad player is detected. iOS 4.2, which will be released in November 2010 will have resolved this issue.

Instructions are displayed below the game area, and a logo above the game. It is also possible to make it a full-screen (1024x768 pixels) game on the iPad, with 1 simple change. This was not done, in order to be able to provide the player with instructions though.

On the iPad version, touch swipe controls are enabled by default. This means that the player can drag the pad with his fingers to any position. Touching anywhere above or below the pad works.

Right now, Android players will see the same game graphics as iPhone players. There are many different Android devices with different screen resolutions. BrickIt supports to change the resolution upon initialization (this is in fact done for 3 versions: PC, mobile and iPad), so it could be adapted to fit the (most popular) Android devices with even higher quality graphics then they will get now.

2.3 Controls

2.3.1 Touch Screen

Tap the left side (left 33%) of the screen to move the pad to the left. Tap the right side (right 33%) of the screen to move the pad to the right. Tap the middle part (middle 33%) of the screen to release the ball.

On an iPad, you can also move the pad by touching the screen on, above or below the pad, and moving your finger. This feature is not enabled for other mobile devices, because it is not very practical on a small screen: the players fingers are constantly obstructing the view. Test users

liked the normal tap/touch controls better. Because it is confusing to have them both enabled at the same time, it is chosen to disable the swipe controls by default on other mobile devices. this can be easily activated though, through a parameter (see Section 4).

To pause, touch the pause icon '||' in the right bottom corner of the game area.

Navigate the menus by touching on the names of menu items. Select a level by touching the triangles. Tap the Sound FX checkbox to enable/disable sound effects.

2.3.2 Mouse

Move the mouse in the game area to move the pad. Click to release the ball.

To pause, click the pause icon '||' in the right bottom corner of the game area.

Navigate the menus by clicking on the names of menu items. Select a level by clicking on the triangles. Click the Sound FX checkbox to enable/disable sound effects.

2.3.3 Keyboard

Move the pad with the left/right arrow keys. Release the ball with the space bar or enter key.

To pause, press escape.

Navigate the menus with the up/down arrow keys. Select a level by navigating to the Level option - from there, use the left/right arrow keys to select. Press 's' to enable/disable Sound FX.

3 Tested Devices

BrickIt is tested on the following mobile devices:

- Apple iPhone 3GS (iOS 4.0)
- Apple iPhone 4 (iOS 4.1)
- Apple iPad (iOS 3.2)
- Apple iPod Touch (2nd generation, iOS 4.1)
- HTC Desire (Android 2.2)
- HTC G1 (Android 1.6)
- Motorola Droid (Android 2.2)
- HTC Droid Incredible (Android 2.2)

Audio is automatically disabled when a mobile device is recognized. This is done because HTML5 audio support is currently limited in the mobile browsers - Android doesn't support it at all, and iOS doesn't properly cache the audio files, resulting in some files not playing in time at random times. This can however be enabled very easily, if desired.

Currently, when an Android device is detected, the same screen size settings are passed to the game as it would be an iPhone. This is not always the best setting, because there are many different Android devices. This is very easy to adapt though.

When a device is not listed, it doesn't mean it's not supported, only that there was no device available for testing.

BrickIt is tested on the following browser/operating systems:

- Apple Safari [4, 5] (Mac OS X)
- Mozilla Firefox 3.6 (Mac OS X, Linux, Windows)
- Google Chrome 7 (Mac OS X, Linux, Windows)
- Opera 10

4 Technical Information

The code of BrickIt is designed with scalability and extensibility in mind. It is name-spaced in the `net.pretopia.BrickIt()` object. It accepts optional parameters to define client-specific aspects like screen size, input devices, audio etc. The web site hosting the game can easily pass this to the game.

No third-party frameworks (like jQuery) were used for the development of this game.

The BrickIt code is all placed in 1 file, `BrickIt.js`. A minimized version by Google Closure Compiler (<http://closure-compiler.appspot.com>) is provided as `BrickIt-min.js`. This file is used on the website, as it is about 60% smaller than the original version (lots of comments, whitespace and long variable names).

It was decided to make the code in the BrickIt object as generic as possible - so device-specific code should be placed externally. BrickIt can be tailored to run on different devices by passing parameters. By separating device-specific code from the core logic, the code could be kept as clean as possible, and scalable for other (new) devices that don't exist yet.

An overview of optional parameters for BrickIt, that are passed to the object as an argument of the `main()` function:

```
/* optional properties in params object
 *
 * name          type          value
 * -----+-----+-----
 * level          <int>         level to start with
 * lives          <int>         number of lives
 * attachKeyboard <bool>        enable keyboard controls for this instance
 * attachTouchControls <bool>    enable touch controls for this instance
 * attachTouchSwipe <bool>     enable touch swipe pad controls for this instance
 * enableSoundFX   <bool>     enable sound effects
 * audio_hitBrick  <string>    id of audio element
 * audio_hitPad    <string>    id of audio element
 * audio_loseLive  <string>    id of audio element
 * audio_startGame <string>    id of audio element
 * audio_finishLevel <string>  id of audio element
 * audio_gameOver  <string>    id of audio element
 * audio_winner    <string>    id of audio element
 * gameAreaWidth   <int>       width of the game area in pixels
 * gameAreaHeight  <int>       height of the game area in pixels
 * brickWidth      <int>       width of brick in pixels
 * brickHeight     <int>       height of brick in pixels
 * padWidth        <int>       width of pad in pixels
 * padHeight       <int>       height of pad in pixels
 * padBottomOffset <int>       offset in pixels of pad from bottom
 * ballRadius      <int>       radius of ball in pixels
 * padSpeed        <int>       speed of pad in pixels/tick
 * defaultBallSpeed <int>       default speed of ball in pixels/tick
 * coreLoopInterval <int>     number of milliseconds between each tick
 */
```

BrickIt can be tweaked to get a higher performance on older (slower) devices, by means of

the `coreLoopInterval` and `defaultBallSpeed` parameters. This defines the number of milliseconds each calculation cycle is supposed to take, as well as the smoothness of the gameplay.

Embedding the game in a page requires only a few lines of code. Example code for a website featuring BrickIt:

```
<html>
  <head>
    <title>BrickIt</title>

    <style media="screen" type="text/css">
      #BrickItDiv {
        margin: 0px auto;
        background: #000000;
        width: 655px;
        height: 455px;
      }
    </style>

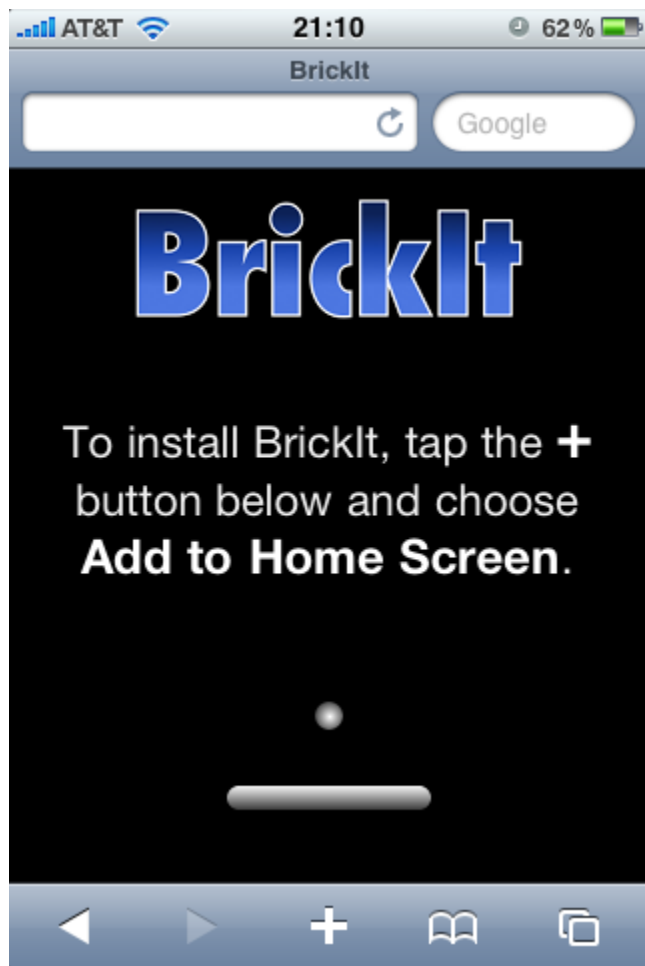
    <script type="text/javascript" src="BrickIt.js"></script>
  </head>

  <body>
    <div id="BrickItDiv"></div>

    <script type="text/javascript">
      /*  */
      var BrickItInstance = new net.pretopia.BrickIt('BrickItDiv');

      var params = {
        // optional parameters
      };

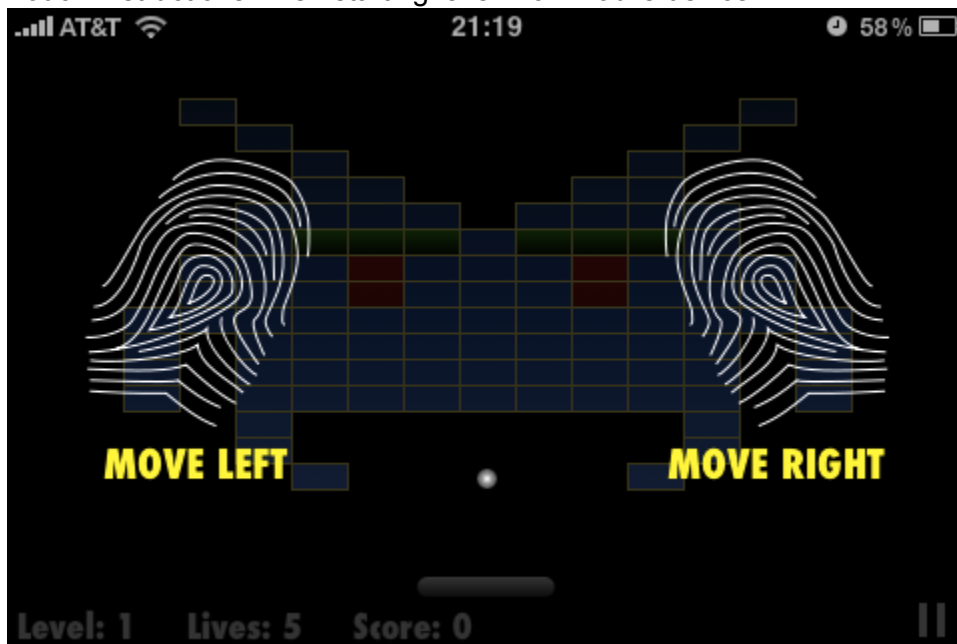
      BrickItInstance.main(params);
      /* ]]]&gt; */
    &lt;/script&gt;
  &lt;/body&gt;
&lt;/html&gt;</pre></div><div data-bbox="111 586 838 621" data-label="Text"><p>It is possible to instantiate multiple instances of BrickIt on one web page (not that there's a reason for wanting to do that, though).</p></div><div data-bbox="111 644 320 667" data-label="Section-Header"><h2>5 Screenshots</h2></div><div data-bbox="111 674 828 708" data-label="Text"><p>Many screenshots needed to be scaled down in order to fit on the pages. Therefore, they provide only an impression of the presentation.</p></div><div data-bbox="111 725 234 745" data-label="Section-Header"><h3>5.1 iPhone</h3></div><div data-bbox="111 751 269 767" data-label="Text"><p>Installation screen:</p></div>
```



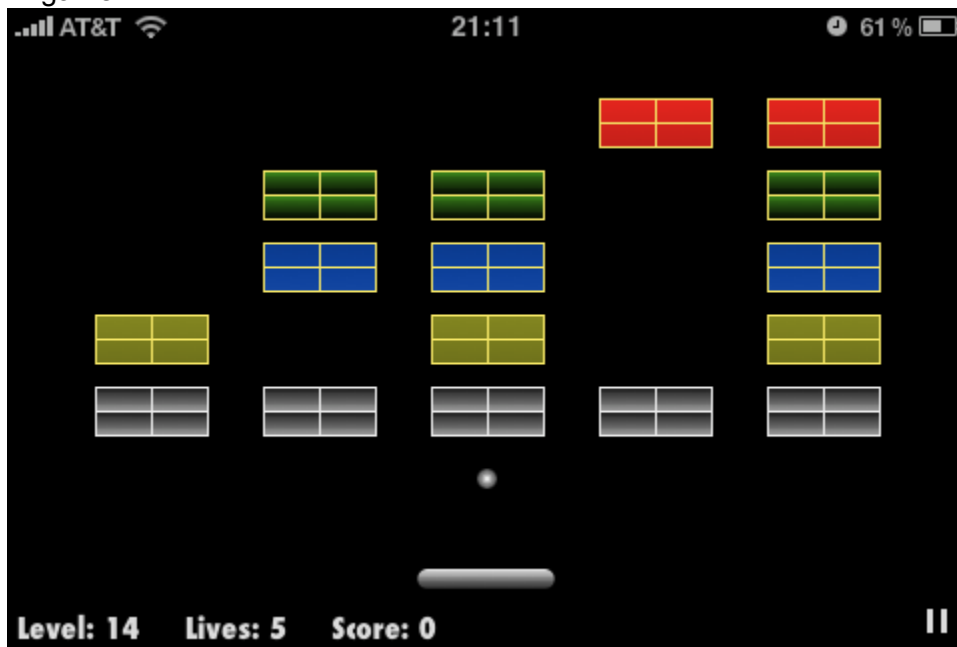
Menu:



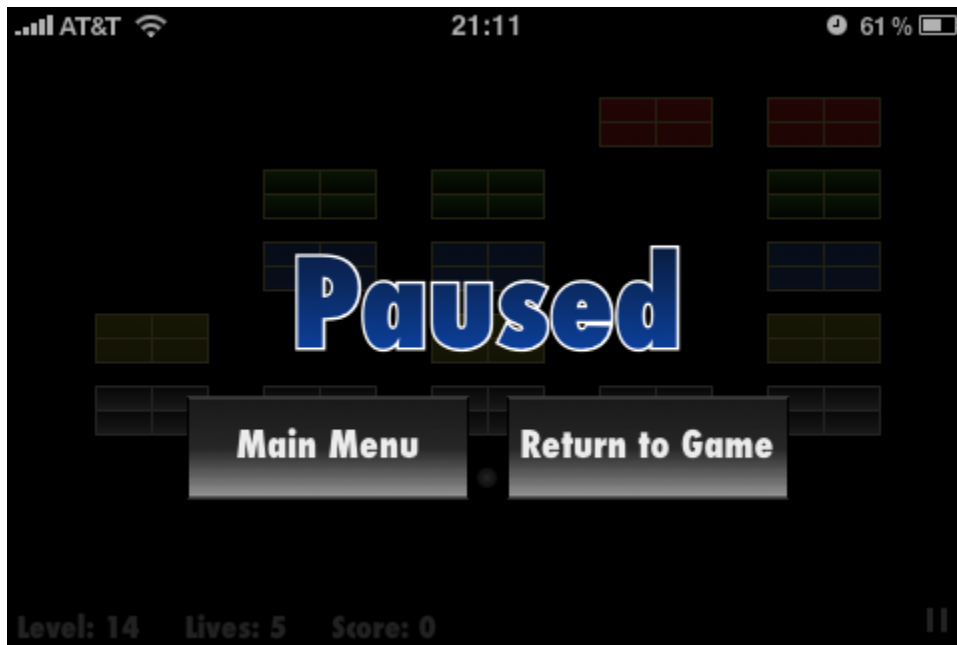
Touch-instructions when starting level 1 on mobile device:



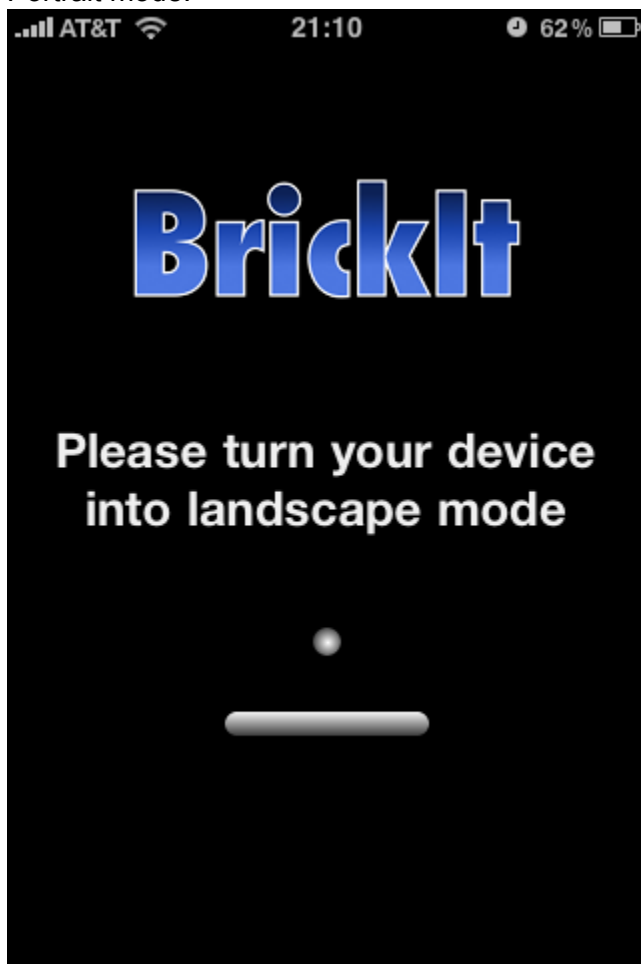
In-game:



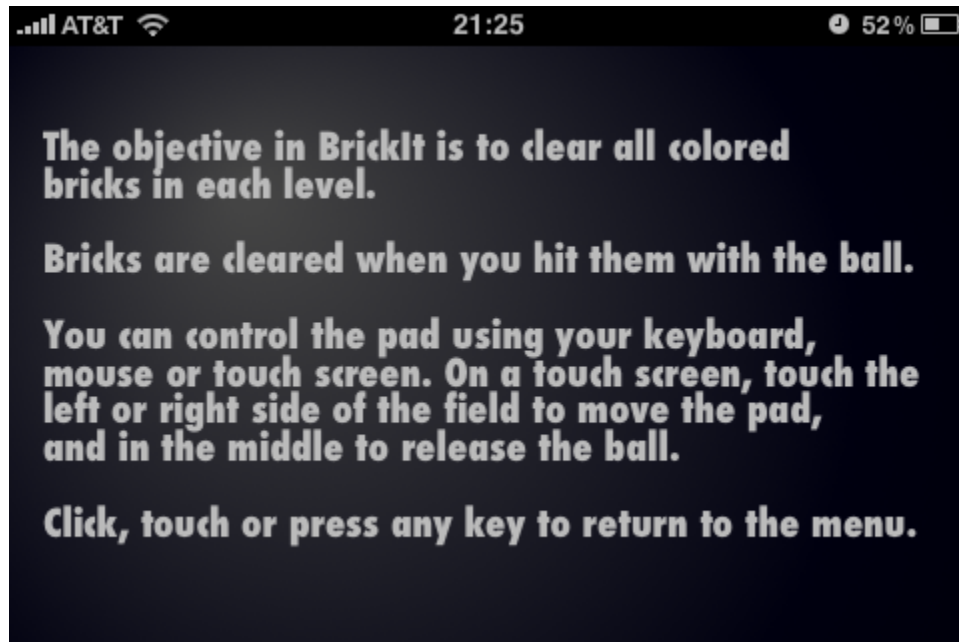
Pause:



Portrait mode:

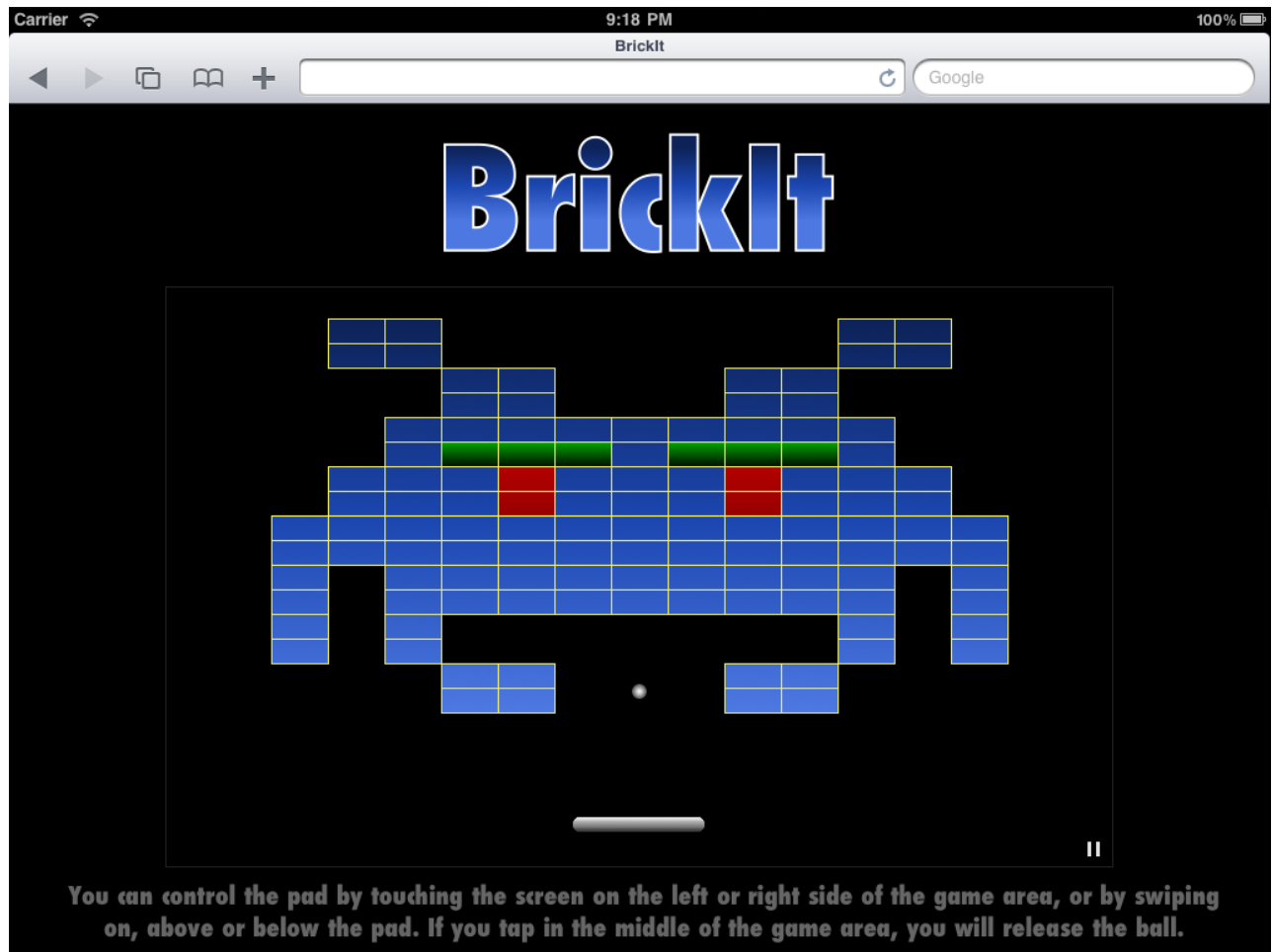


Instructions:



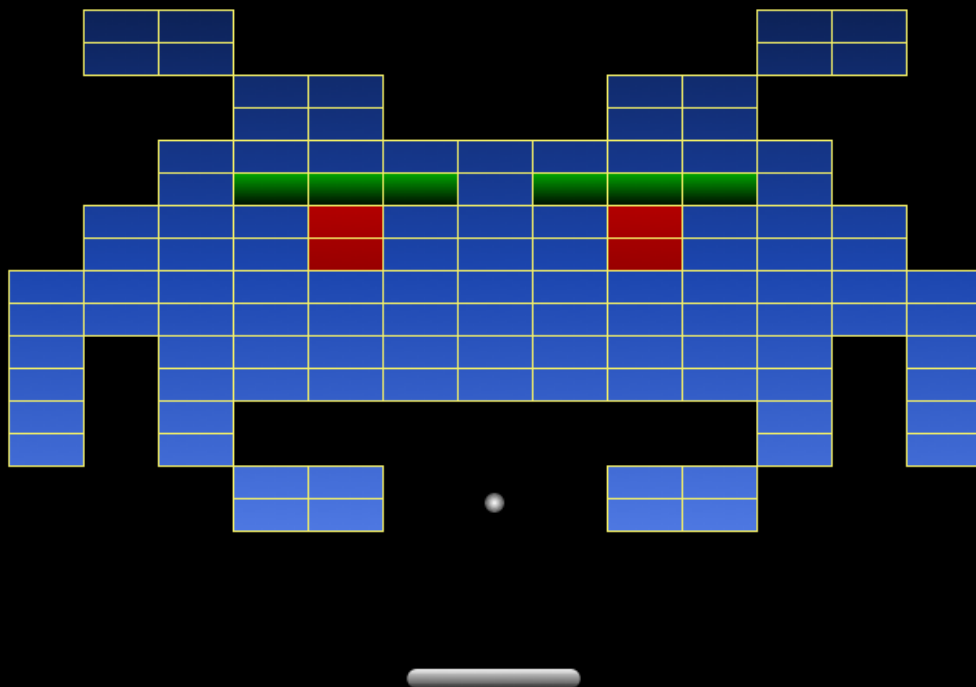
5.2 iPad

Landscape mode:



Portrait mode:

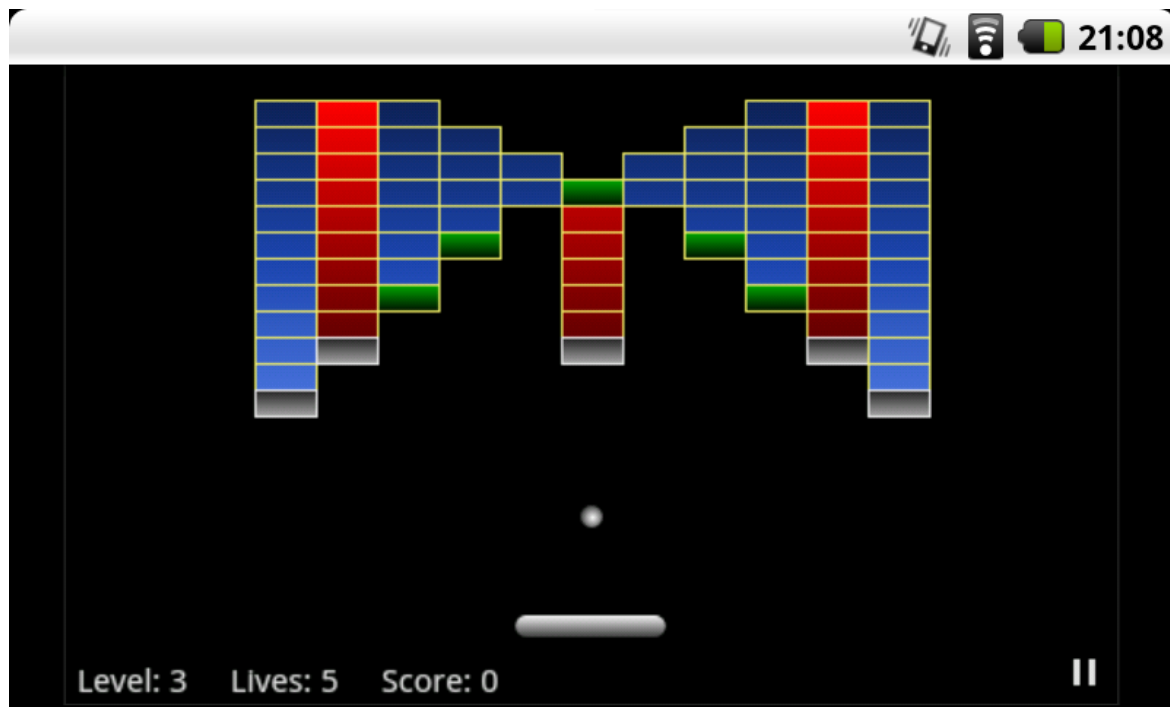
BrickIt



You can control the pad by touching the screen on the left or right side of the game area, or by swiping on, above or below the pad. If you tap in the middle of the game area, you will release the ball.

5.3 Android

In-game:

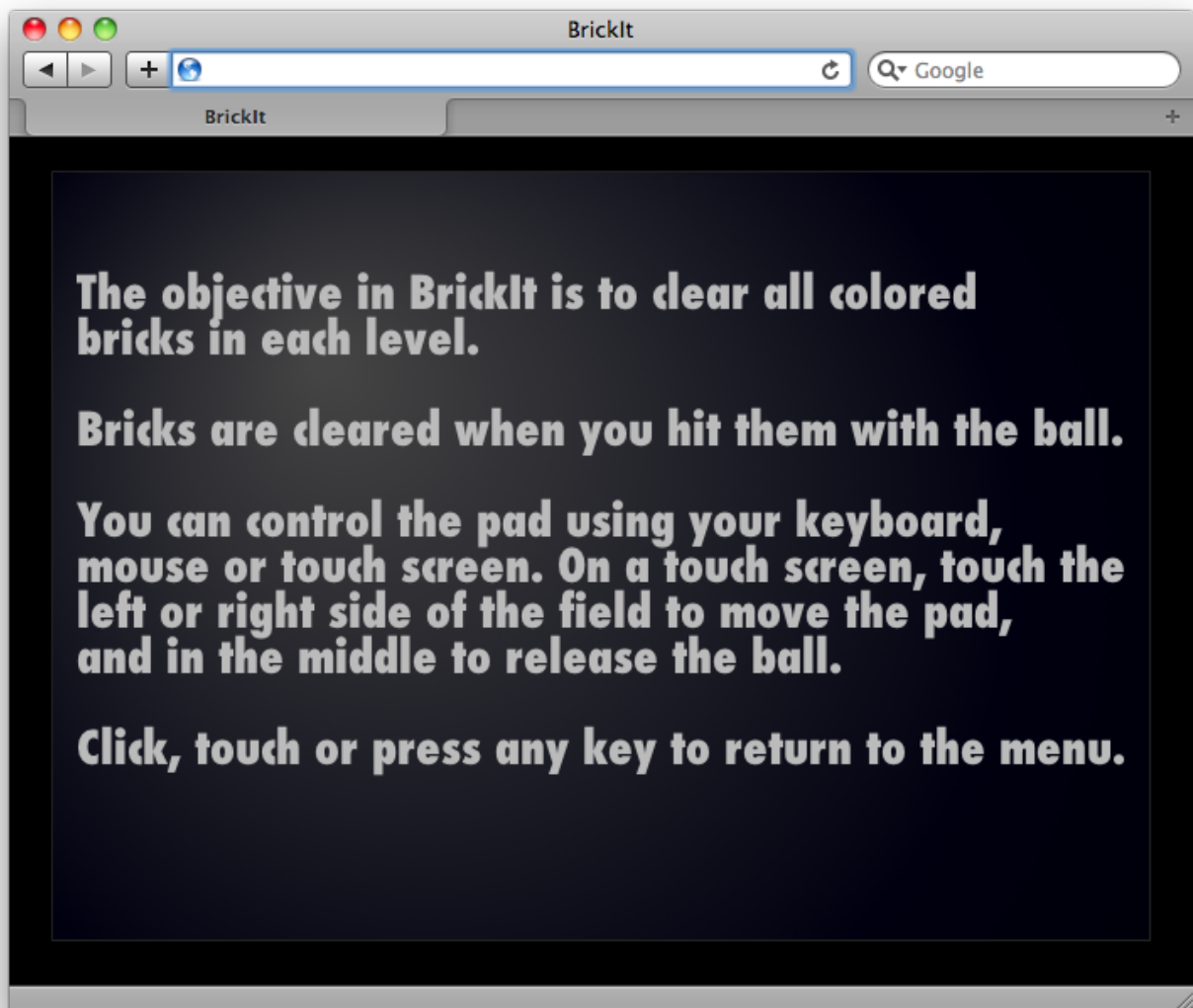


5.4 PC Browser

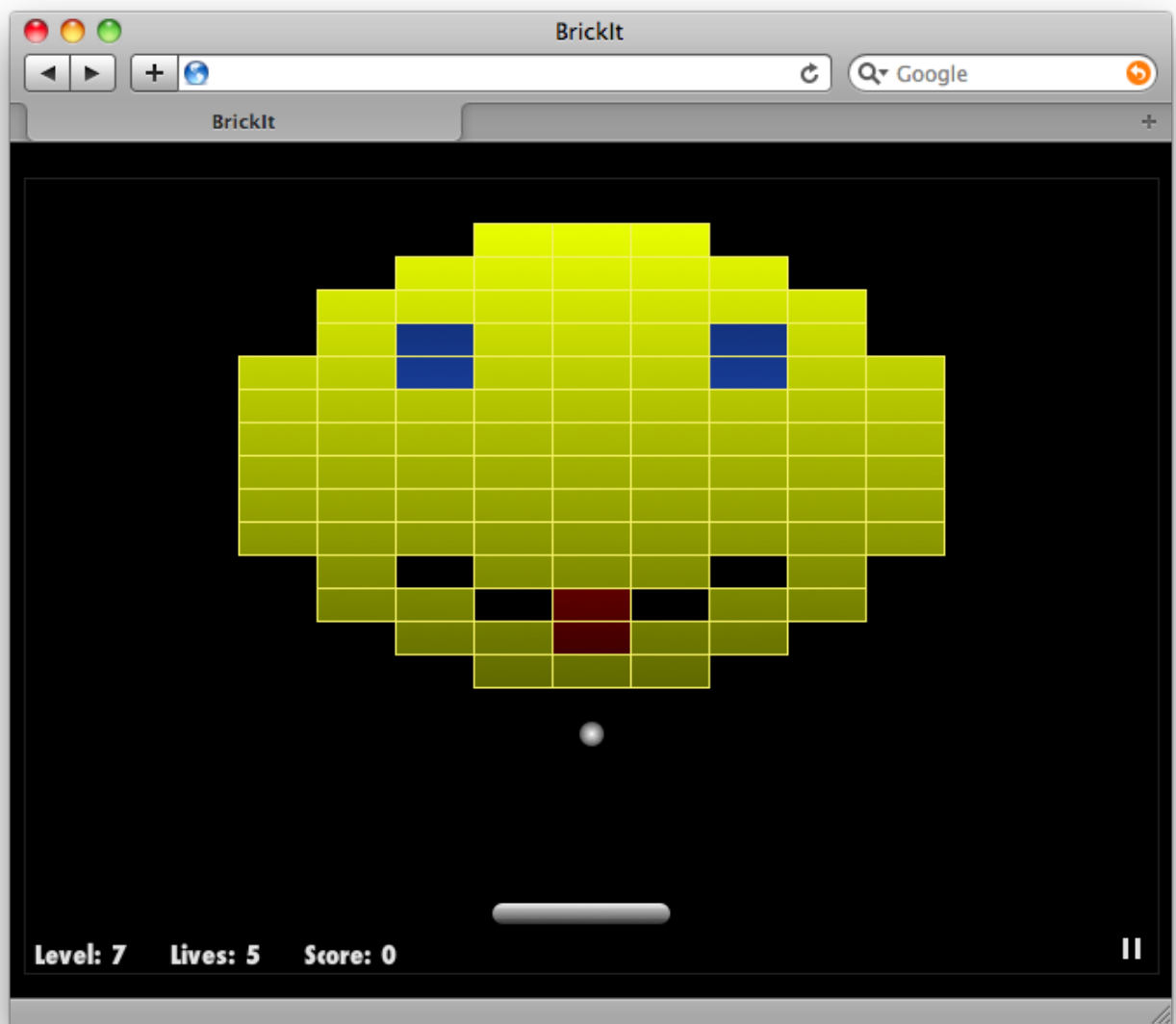
Menu (the ball means that keyboard input is activated - it functions as a cursor):



Instructions screen:



In-
game:



Pause (the white border around 'Return to Game' shows keyboard focus):

